



Entra the Matrix Where Security Copilot is your Neo!



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Security Technical Specialist | CCSP CISSP CISM

The Microsoft Copilot logo, a stylized, three-dimensional knot-like shape in shades of blue and grey, positioned above the text "MICROSOFT COPILOT".

MICROSOFT
COPILOT



A portrait of Keanu Reeves as Neo from the movie The Matrix, wearing his signature black sunglasses and a black turtleneck. He is centered in the background of the slide.

Agenda

Introduction Security Copilot
Explore Security Copilot in Entra
Unlock hidden identity features

MICROSOFT
COPILOT

The Microsoft AI Landscape



AI Elements for
App Dev

Microsoft
Copilot

Consumer

Commercial

for
Dynamics
365

for
Fabric

for
Microsoft
365

for
Sales

GitHub
Copilot

in Power
Platform

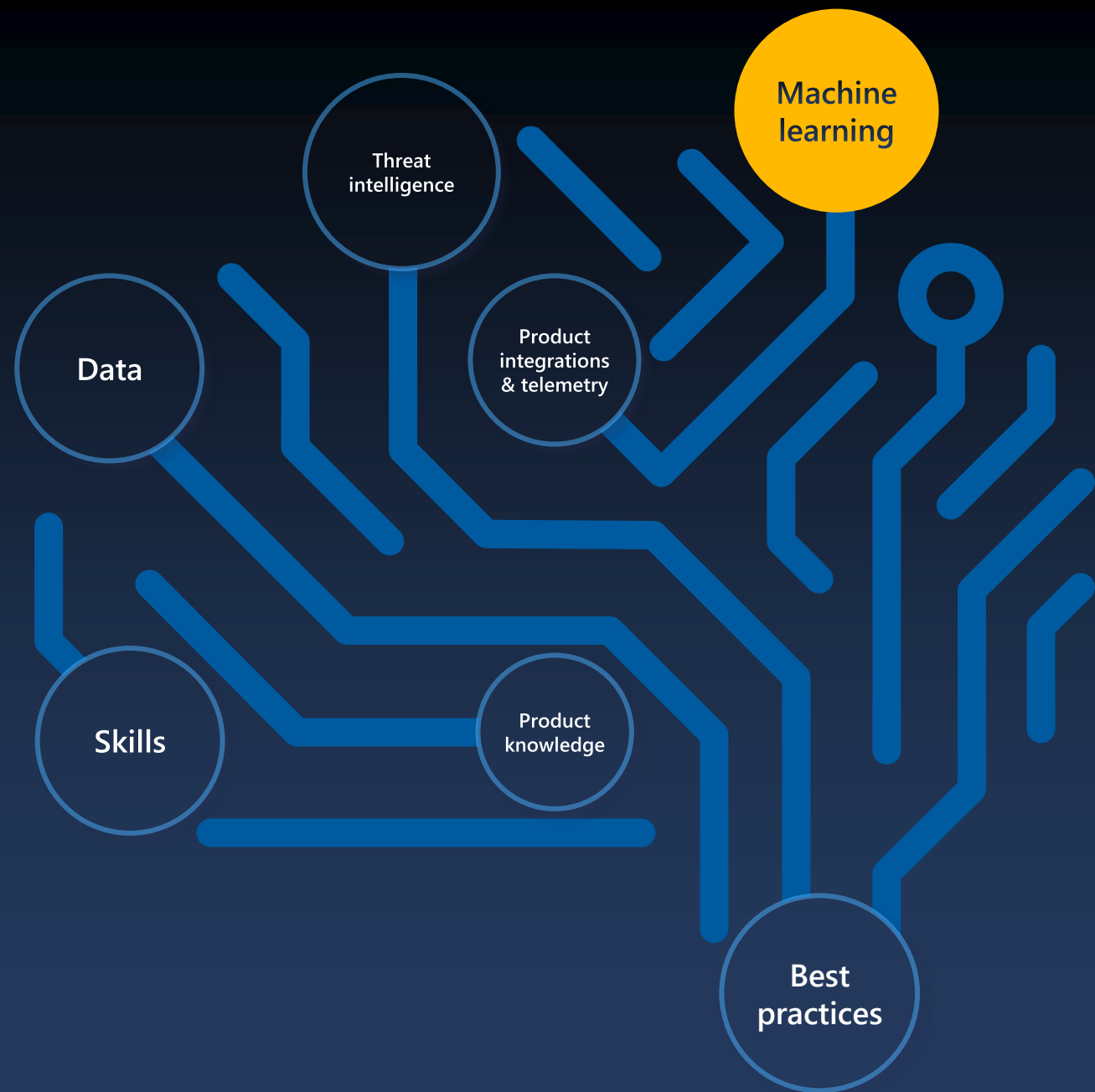
in
Windows

for
Security



Microsoft Security Copilot

The first generative AI security product that empowers security and IT teams to defend at the speed and scale of AI



End-to-end security at machine speed and scale

Microsoft Security Solutions

Available in the
standalone experience

Available as an
embedded experience



Microsoft
Defender XDR



Microsoft
Sentinel



Microsoft
Intune



Microsoft
Entra



Microsoft
Purview



Microsoft
Defender for
Cloud



Rapid investigation and response

Investigate with AI-assisted insights and quickly pivot to remediation with actionable, prioritized recommendations

Scaled visibility

Quickly assess security posture, threats and policy or compliance gaps. Access summaries with context to understand the potential impacts.

Faster troubleshooting

Get deep understanding of device, user, access, and app status to resolve issues quickly. Find and remediate policy issues faster with natural language prompts.

Advanced skills unlocked

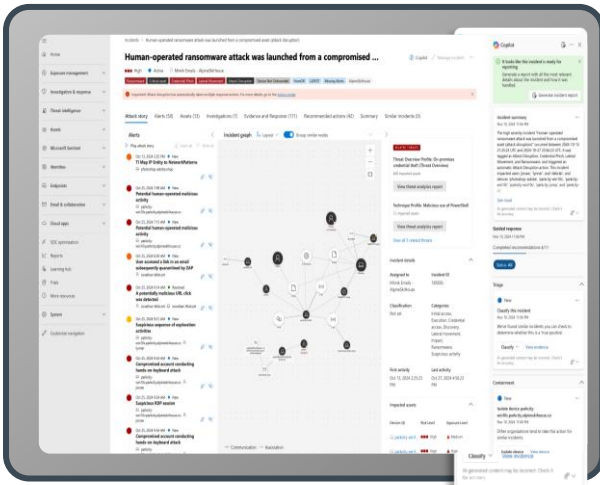
Script analysis and natural language to KQL and KeyQL empower any team member to complete complex tasks with confidence.

*Available as part of the Unified Security Operations Platform.

Evolution of Security Copilot

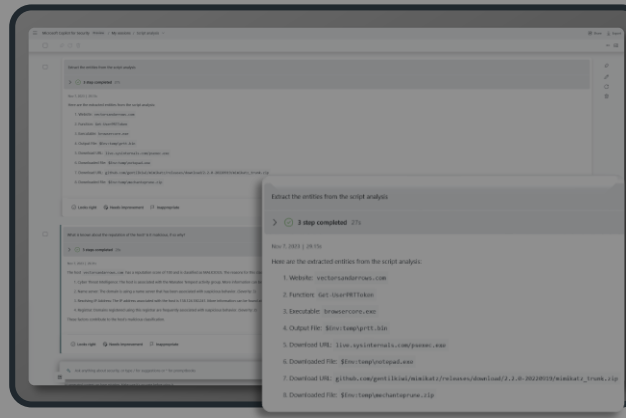
Embedded

Offers the intuitive experience of getting Copilot guidance natively within the products that your team members already work from and are familiar with



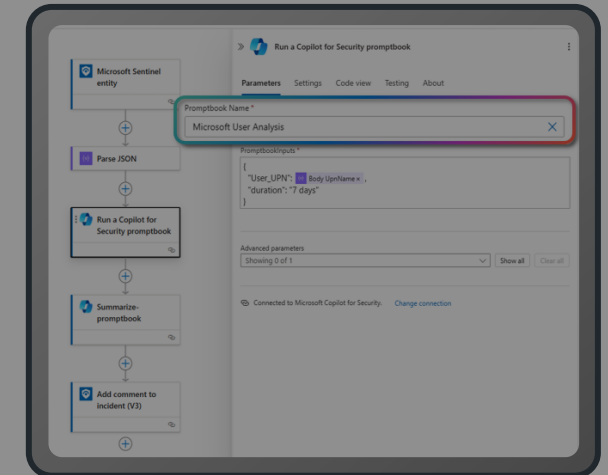
Standalone

Helps teams gain a broader context to troubleshoot and remediate incidents faster within Copilot itself, with many use cases in one place, enabling enriched cross-product guidance



Automation

Helps teams accelerate response with built-in and custom promptbooks as well as integration with Logic Apps



Security Copilot Platform

Demo #1

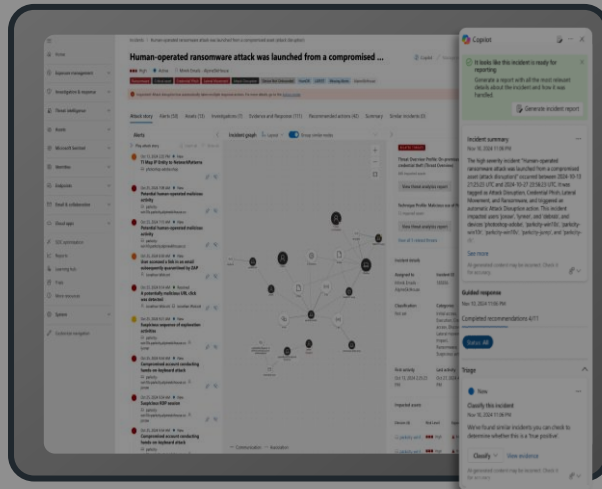
Setup Security Copilot & integration
in Microsoft Entra

MICROSOFT
COPILOT

Evolution of Security Copilot

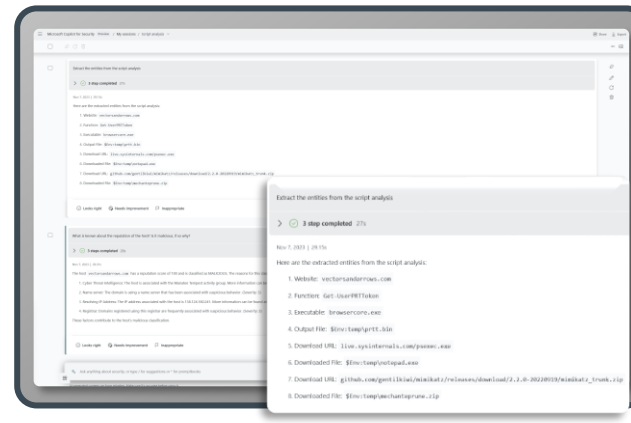
Embedded

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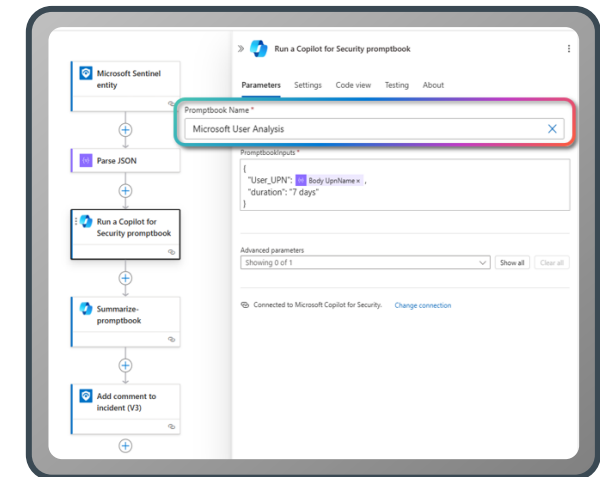
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Automation

Helps teams accelerate response with built-in and custom promptbooks as well as integration with Logic Apps



Security Copilot Platform

Elements of effective prompts

Goal

What is the specific security-related information you need?

+

Context

What is the workflow focus or analysis for a tailored response?

+

Source

Is there a plugin, known info, or data source Security Copilot should use?

+

Expectations

How will the information be used and what is the desired format?

"Give me information about Simon Templar..."

"...that includes the most recent risky sign-ins..."

"...in Entra..."

"...summarize into a list of recommendations that I can submit to my manager."

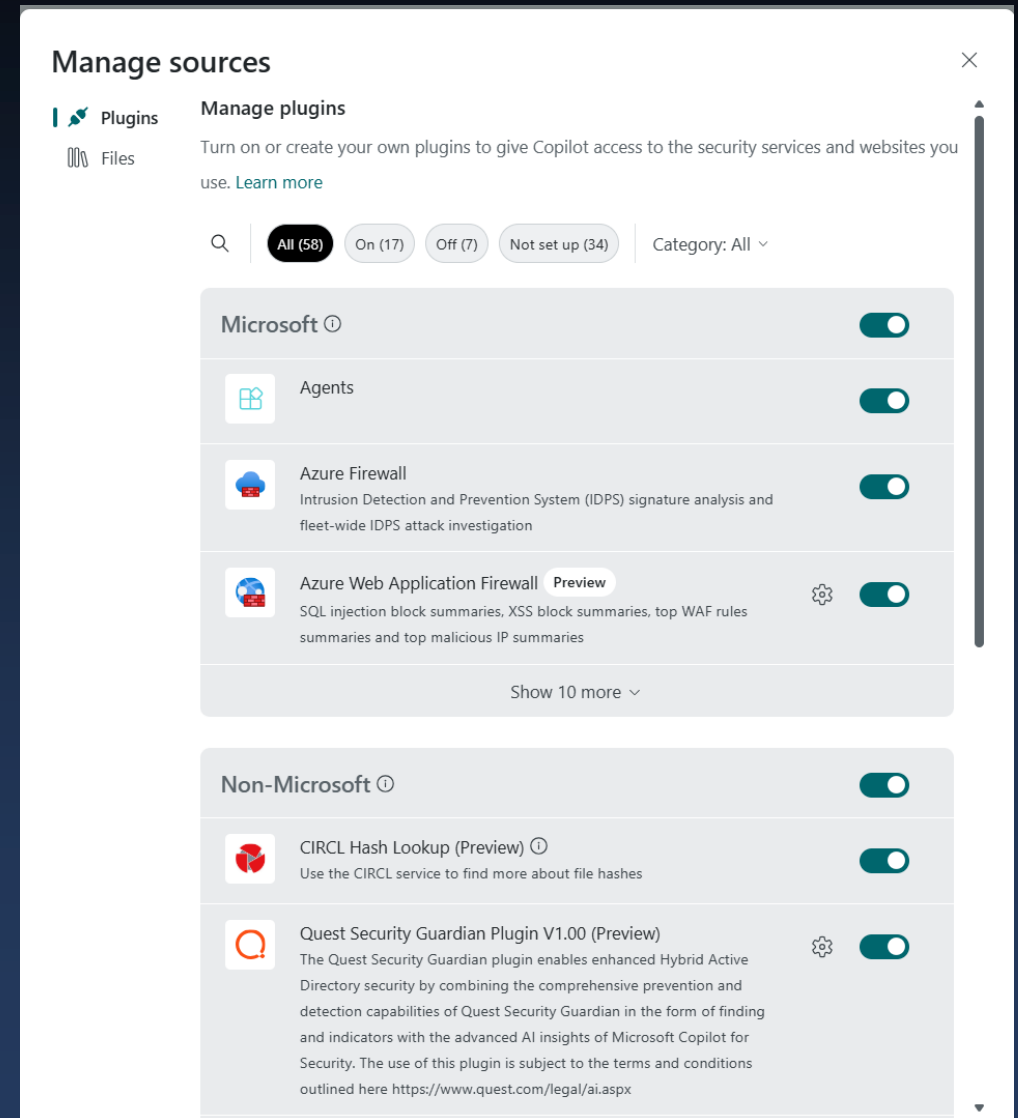
Plugins

Pre-built

Pre-built plugins from 1st & 3rd Party applications & services. Invoke skills within plugins using natural-language prompts or direct skill invocation. Enable Plugins for all users or just you; Admins have the control!

Custom

You can build your own plugins (API, GPT, KQL) to enrich your investigations.



Custom plugin examples



API plugins

- › Additional threat intelligence sources
- › Device IoT systems
- › Perimeter defense systems



KQL plugins

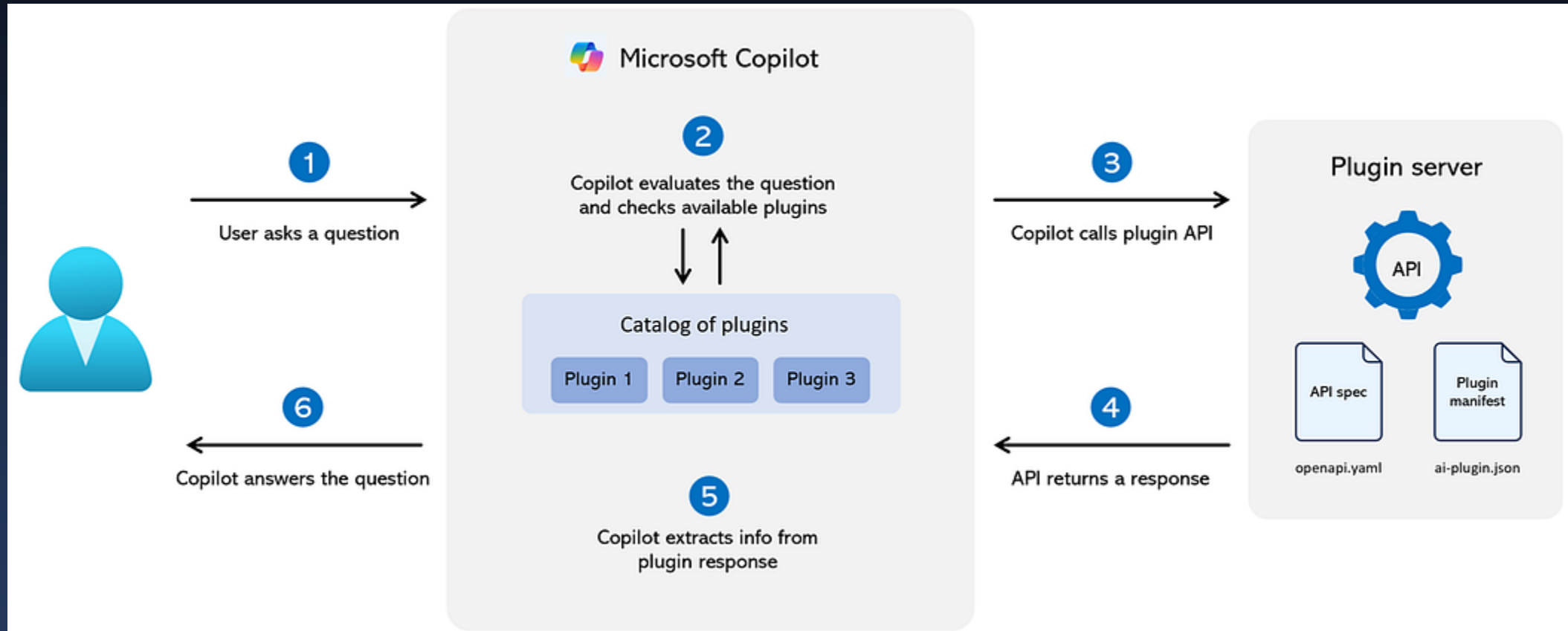
- › Query a KQL database
- › Log analytics workspaces
- › Defender XDR Advanced Hunting



GPT plugins

- › Summarize sessions in specific formats
- › Define reporting structure
- › Reformat specific subtext

How to interact with Security Copilot



A plugin consists of two main components

OpenAPI Specification

This component describes how the API of the plugin works. It defines the endpoints, request and response formats, authentication mechanisms, and any other relevant details about the plugin's API.

```
openapi: 3.0.0
info:
  title: Microsoft Graph API Plugin to access data from Entra ID and Conditional Access Policies
  description: Specification to retrieve information from the Graph API for Conditional Access Policies
  version: "1.0"
servers:
  - url: https://graph.microsoft.com/v1.0
paths:
  /identity/conditionalAccess/policies:
    get:
      operationId: ListCAPolicies
      summary: List all Entra ID Conditional Access Policies.
      responses:
        200:
          description: Successful retrieval.
        401:
          description: Unsuccessful authentication.
```

Plugin Manifest

The plugin manifest file is used to explain to Copilot how to use the plugin. The manifest file helps Copilot understand how to integrate and interact with the plugin.

```
Descriptor:
  Name: Conditional Access - Graph API Plugin
  DisplayName: Conditional Access - Graph API Plugin
  Description: Retrieves information from Microsoft Graph API with the current user permissions.
  Authorization:
    Type: AADDelegated
    EntraScopes: https://graph.microsoft.com/.default
SkillGroups:
  - Format: API
    Settings:
      OpenApiSpecUrl: https://raw.githubusercontent.com/rdejong1979/security-copilot-plugins/refs/heads/main/identity/conditionalAccess/policies/openapi.yaml
```

KQL Type Plugin

Help the orchestrator to select this plugin with good description and example prompts

Descriptor:
Description: >
Identity Security skills to query Identity events for detection and forensics hunting across User Risk Assessment, Sign-in Monitoring, Admin Activity Monitoring, Application Usage Monitoring, Privileged Identity Management, Access Review

SkillGroups:
- Format: KQL
Skills:
- Name: IdentityGetUserRiskAssesment
DisplayName: Get User Risk Assessment
Description: >
Fetches the user risk levels based on their activities. This could include sign-in attempts from unfamiliar locations, repeated failed sign-in attempts, or other suspicious behavior.
ExamplePrompt:
- "Get all risky users"
- "Fetch risky users"
- "Run users risk assessment"
- "List all risky users"

Target can be:

- Sentinel
- Defender
- Log analytics
- ADX

Settings:
Target: Defender //|
Template: |-
// Query to fetch user risk events with sign-in details
let RiskyEvents = AADUserRiskEvents
| where TimeGenerated > ago(1d)
| project UserPrincipalName, RiskLevel, RiskEventType, TimeGenerated;
let SignInAttempts = SigninLogs
| where TimeGenerated > ago(1d) // Same time filter
| summarize AttemptCount = count(), FailedAttempts = sumif(1, ResultType != 0) by UserPrincipalName, Location = toString(LocationDetails), IPAddress, TimeGenerated;
RiskyEvents
| join kind=inner (SignInAttempts) on UserPrincipalName
| where RiskLevel != "None" // Filter out non-risky users
| project UserPrincipalName, RiskLevel, RiskEventType, AttemptCount, FailedAttempts, Location, IPAddress, TimeGenerated
| order by RiskLevel desc, TimeGenerated desc
| take 1000

Query
input

Return only the fields
you need

```
1 openapi: 3.0.1
```

```
2 info:
```

```
3   title: Microsoft Graph API - List Conditional Access Policies
```

```
4   description: Retrieve a list of conditional access policy objects.
```

```
5   version: 1.0.0
```

```
6 servers:
```

```
7   - url: https://graph.microsoft.com/v1.0
```

```
8   path:
```

```
9     /identity/conditionalAccess/policies:
```

```
10    get:
```

```
11      summary: List conditional access policies
```

```
12      description: Retrieve a list of conditional access policy objects.
```

```
13      operationId: listConditionalAccessPolicies
```

```
14      parameters:
```

```
15        - name: $skip
```

```
16          in: query
```

```
17          description: Skip the first n results.
```

```
18          required: false
```

```
19          schema:
```

```
20            type: integer
```

```
21        - name: $top
```

```
22          in: query
```

```
23          description: Show only the first n results.
```

```
24          required: false
```

```
25          schema:
```

```
26            type: integer
```

```
27        - name: $count
```

```
28          in: query
```

```
29          description: Include count of items.
```

```
30          required: false
```

```
31          schema:
```

```
32            type: boolean
```

```
33        - name: $filter
```

```
34          in: query
```

```
35          description: Filter items by property values.
```

```
36          required: false
```

```
37          schema:
```

```
38            type: string
```

```
39        - name: $orderby
```

```
40          in: query
```

```
41          description: Order items by property values.
```

```
42          required: false
```

In the **info object** is like a summary of key details about the API.

It usually includes things like the title, a brief description, the version number, and links to stuff like licenses and terms of service.

The only things that are required in there are the title and version

In the **servers object**, we can list one or more main paths used in requests to the API.

This section is in array format, so each path needs to be specified separately, unlike the values in the info section.

The basepath is the part of the web address that comes before the specific endpoint. The only required property is the url

The Paths object

The paths object can be seen as a roadmap for the endpoints available within the API.

It provided details on how to access the endpoints and what actions you can perform.

Each endpoint is represented by a unique path within the paths object

To break this section a bit more down into digestible chunks, I will describe the most important parts.

- path
- HTTP methods
- operation object ...

Operation Object

For each HTTP method specified, there must be an associated operation object containing additional information about the operation.

Examples of these operation objects are parameters, request body, responses, etc.

The only required values are summary and the operation object responses

The summary field provides a brief, human-readable description of the endpoint.

In the responses section, we define the possible responses that the API can return when that endpoint is accessed. It contains the HTTP status codes and their corresponding descriptions.

Optionally, the responses section includes details about the response payload, like the format and schema definition.

Parameter Object

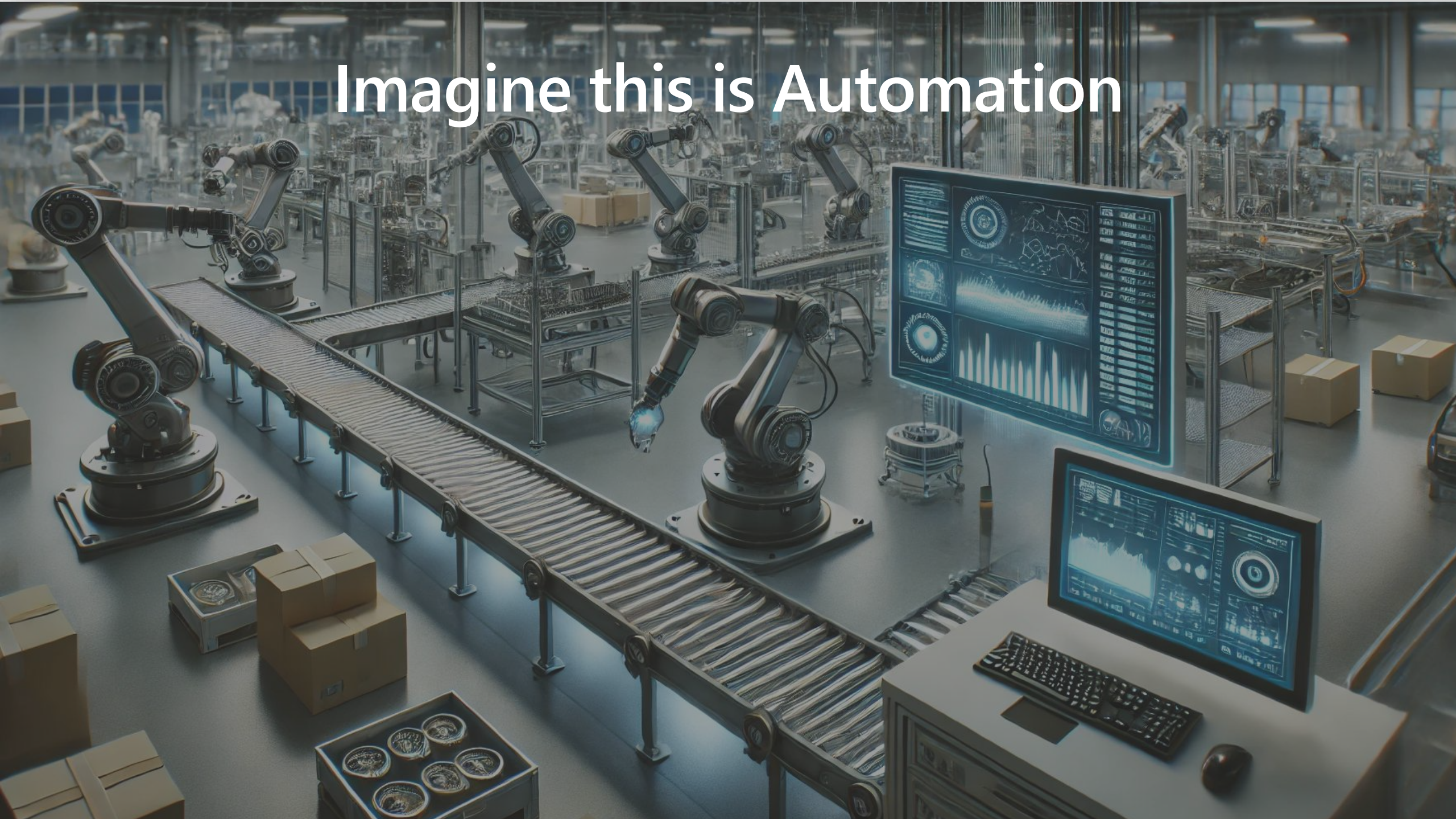
You don't have to use the parameter object, but it's handy when making a plugin for Copilot(s).

If you include the parameter object in the path item, the parameters will affect all actions on that path.

Alternatively, you can place parameters in the operations object, where they'll only affect that specific action.

Depending on the API, the parameters can reside in different locations, indicated by the in field.

Imagine this is Automation



Automation is characterized by

Task nature

Predefined repetitive rules-based tasks.

Adaptability and learning

Low. Unexpected/new scenarios require human intervention.

Task complexity

Best suited for repetitive linear tasks or trigger workflows in response to triggers.

Demo #2

Automation & extend Security Copilot
with Microsoft Entra

MICROSOFT
COPILOT

Then this is an agent



An agent is different

Task nature

Simulates intelligent behavior to create and plan dynamic workflows.

Adaptability and learning

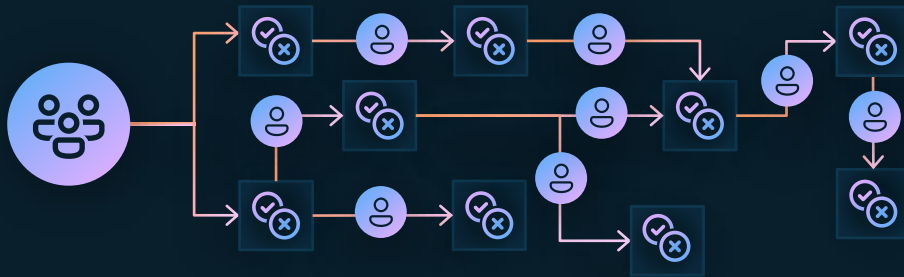
High. Agents learn, analyze patterns, and adapt using reinforcement learning.

Task complexity

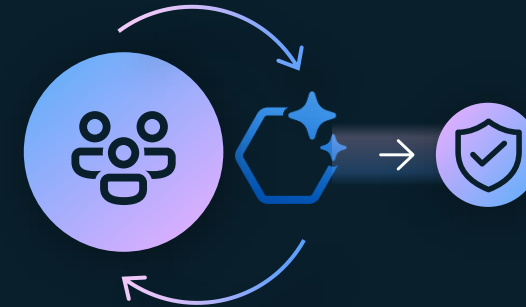
Well-suited for complex, non-linear tasks requiring decision making.

Agents go further than traditional automation

Traditional automation



Agentic AI



Rigid



Adaptive

Static



Dynamic

Manual updates



Continuous learning

Pre-defined



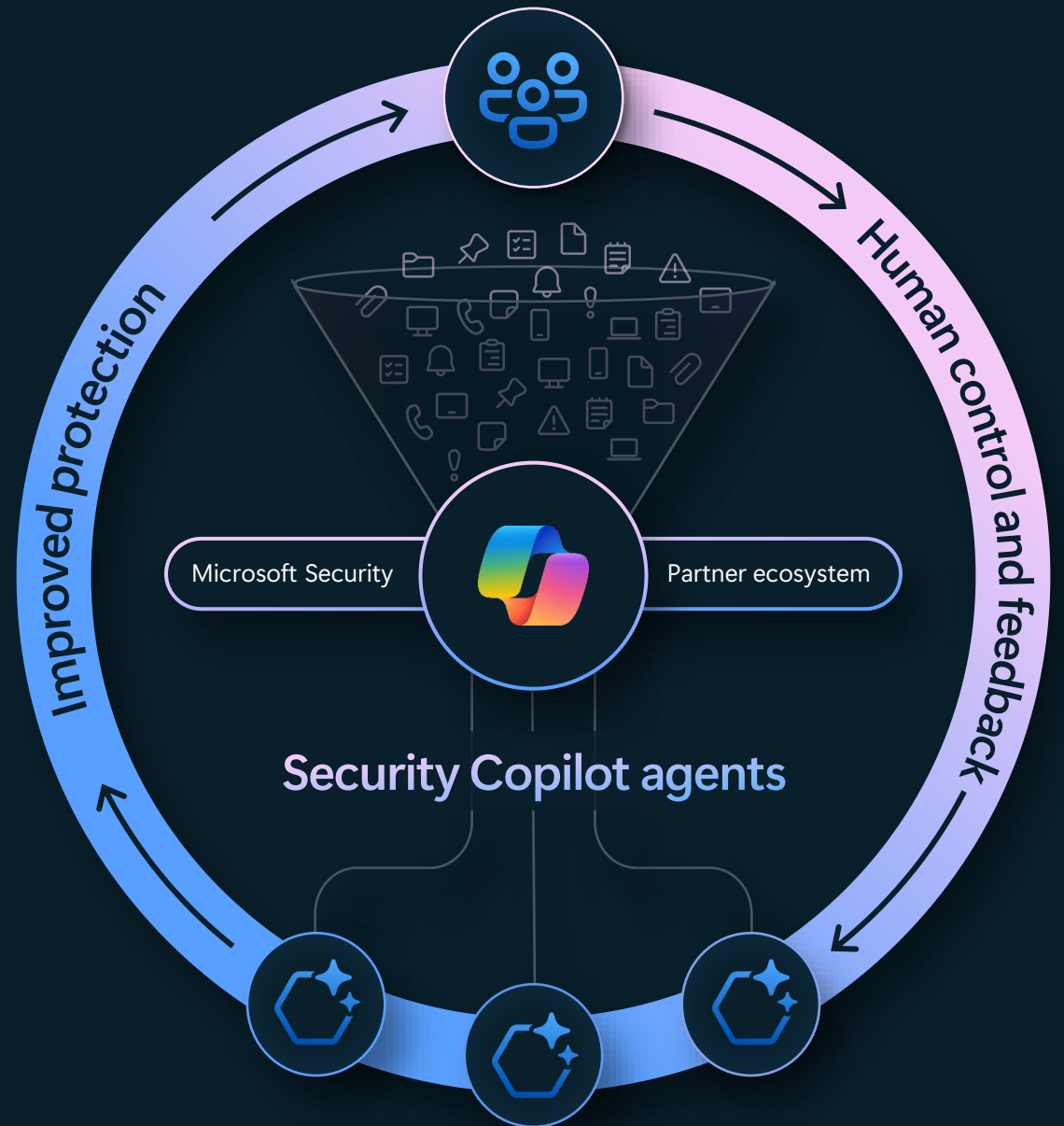
Context-aware

Smarter defense with Security Copilot agents

Seamlessly integrated with Microsoft Security solutions and partner ecosystem, **Microsoft Security Copilot agents** enhance security and IT operations with autonomous and adaptive automation.

Purpose-built for security, agents:

- ✓ continuously learn from feedback
- ✓ dynamically reason and adapt with your team fully in-control
- ✓ operate securely within Microsoft's Zero-Trust framework



Conditional Access Optimization Agent in Microsoft Entra

Quickly identify security gaps

The agent continuously monitors for new users and apps to detect misalignments with Conditional Access policies, reducing unnoticed vulnerabilities.

One-click fixes

Get actionable recommendations with easy one-click remediation, streamlining CA policy updates and enhancing security with minimal effort.

Secure access as conditions change

The agent continually responds to changes in the environment, enhancing protection and reducing manual audits.

Microsoft Entra admin center

Search resources, services and docs

Copilot

Connie Wilson

Home > Agents >

Conditional Access Optimization Agent (Preview)

Microsoft Entra ID

Start agentStop agentRemove agentGive Microsoft feedback

OverviewSuggestionsActivitySettings

The agent completed its run

Review any suggestions about policy updates and new policies

Agent is active

Agent finished running on March 24, 2025 at 8:15 AM. View run

The agent is next scheduled to run on March 25, 2025 at 8:00 AM.

Recent suggestions

AI-generated content may be incorrect. Check it for accuracy

View the agent's suggestions about policy updates and new policies that were created in report-only mode.

Suggested next stepsActions taken by agentScopeCoverageTime generated

Recent suggestions

AI-generated content may be incorrect. Check it for accuracy

View the agent's suggestions about policy updates and new policies that were created in report-only mode.

Suggested next steps	Actions taken by agent	Scope	Coverage
Add 16 users to existing policy	Suggested policy update	Users	MFA policy
Turn on new report-only policy for 32 users	Created new policy	Users	Device compliance policy
Add 28 users to existing policy	Suggested policy update	Users	Suggested policy update
Add 9 users to existing policy	Suggested policy update	Users	Device compliance policy

Demo #3

Conditional Access Optimization Agent
in Microsoft Entra

MICROSOFT
COPILOT

Microsoft Entra admin center

Search resources, services, and docs (G+/)

Copilot

Connie Wilson
CONTOSO OUTDOORS

Home

Agents

Favorites

Entra ID

ID Protection

ID governance

Verified ID

Global Secure Access

Permissions Management

What's new

Billing

Diagnose & solve problems

New support request

Microsoft Entra | Home

Security Copilot agents are here

Discover a whole new way to automate security with AI.

Go to agents

Contoso Outdoors

Tenant ID 485739sjdije03-8jh8-323o4578...

Primary domain contosooutdoors.ms

67,876

View users

56,745

View devices

7,046

View groups

32,423

View apps

Connie Wilson

23904732-238u43u2r32u-32iu4434

View user profile

My Entra role assignments

125 role assignments

Active role assignments

Eligible role assignments

View my roles

55 users at high risk

Number of risky users with risk level "high".

Learn more

5% in the last 30 days

View high risk users

Shortcuts

Add

Sign-ins

Audit logs

Authentication methods

Blocked users

Domain names

Unused service principals

Named locations

Cross-Tenant Access Policies

Tenant restrictions

Risky sign-ins

Risk policies

Get the most out of your licenses and subscriptions

Licenses

Entra P2

Workload ID

Subscriptions

2

View license usage

Deployment suggestions

Microsoft suggests that you deploy these features to improve your security posture

Tasks	Time	Progress
Set up risk-based Conditional Access policies	5 minutes	Not started
Get started with passwordless authentication	8 minutes	Not started
Create an access package for guest users	1 minute	Not started

Tenant status

Identity Secure Score

65.15%

View recommendations

Microsoft Entra Connect

Enabled

View Entra Connect

nt
rosoft
ailable

ls new users and apps in your tenant
e they're covered by policies.

Conditional Access Optimization Agent (Preview) ×



The agent will scan all newly created users and apps from the last 24 hours and determine if they are in scope of at least one policy with multifactor authentication (MFA) or device compliance controls. It will suggest policy changes based on [Zero Trust best practices](#) and create new policies in report-only mode. It won't add new users to existing Conditional Access policies or turn on new policies without your approval. [Learn more about this agent](#)

Trigger

The agent will run once every 24 hours. You will also be able to run the agent manually at any time.

Permissions

- ✓ Read access for users, devices, and applications
- ✓ Read and write access for groups and Conditional Access policies

Identity

The agent will run using your identity. Agent authentication will expire according to your policies and need to be renewed. [Learn more about agent authentication](#)

Products



Conditional Access

Plugins



Microsoft Entra

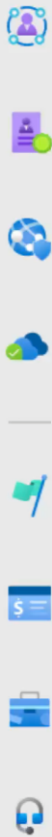
Role-based access

Conditional Access Administrators, Security Administrators, Global Administrators. [Learn more about who can use the agent](#)

Once started, the agent will run in your tenant. You can remove the agent or customize the agent output.

Cancel

Start agent



Generating suggestions for gaps identified ...

 [Stop the agent](#) if you want to stop this run.

 Agent is running

Generating suggestions for gaps identified ...

- ✓ Identified gaps based on data analysis
- ✓ Cross-referenced new users with existing policies to determine if new users are included or excluded
- ✓ Scanned your tenant's Conditional Access policies for policies enforcing multifactor authentication access control
- ✓ Scanned your tenant's audit log to identify new users added in the last 24 hours

Performance highlights ⓘ

Unprotected users discovered ⓘ Sign-ins protected ⓘ

Recent suggestions

AI-generated content may be incorrect. Check it for accuracy

View the agent's suggestions about policy updates and new policies that



The agent completed its run

[Review any suggestions](#) about policy updates and new policies

Agent is active

Agent finished running on March 18, 2025 at 9:07 AM. [View run](#)

The agent is next scheduled to run on March 19, 2025 at 9:00 AM.

Performance highlights ⓘ

Unprotected users discovered ⓘ

63

Sign-ins protected ⓘ

0

About this agent

The agent will scan all newly created users and apps from the last 24 hours and determine if they are in scope of at least one policy with

Recent suggestions

AI-generated content may be incorrect. Check it for accuracy

View the agent's suggestions about policy updates and new policies that

Suggested next steps	Actions
Add 16 users to existing policy	Suggest
Turn on new report-only policy for 47 users	Created

View all

Microsoft Entra admin center

Search resources, services, and docs (G+)

Copilot

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CONTOSO OUTDOORS

Home > Agents >

Conditional Access Optimization Agent (Preview)

Microsoft Entra ID

Start agentStop agentRemove agentGive Microsoft feedback

OverviewSuggestionsActivitySettings

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Unprotected users discovered63

Sign-ins protected0

About this agent

The agent will scan all newly created users and apps from the last 24 hours and determine if they are in scope of at least one policy with multifactor authentication (MFA) or device compliance controls. It will suggest policy changes based on Zero Trust best practices and create new policies in report-only mode. It won't add new users to existing Conditional Access policies or turn on new policies without your approval. Learn more about this agent

Products

Conditional Access

Recent suggestions

View the agent's suggestions about policy updates and new policies that were created in

Suggested next steps	Actions taken by agent
Add 16 users to existing policy	Suggested policy update
Turn on new report-only policy for 47 users	Created new policy

View all

CA3: MFA for Engineering

Conditional Access policy

Policy detailsPolicy impact (Preview)

EditDuplicateDownload JSONDelete

Agent suggestion: Add 16 users to policy

Conditional Access Optimization Agent

The agent found 16 new users that are not in scope of any Conditional Access policies requiring multifactor authentication (MFA) control. The agent has created a new group for those users and has identified the policy CA3: MFA for Engineering as an existing policy that the group can be added to. This policy require users to satisfy MFA controls before accessing resources targeted in the policy. These new users were added to the Contoso tenant by Lauren Baker on 2/22/25 at 1:56 PM.

Apply suggestionReview policy changes

View agent's full activity

AI-generated content may be incorrect. Check it for accuracy

Policy details

Name

CA3: MFA for Engineering

State

Enabled

Edit

Included identities (1)

1 group

View all

Included cloud apps (1)

All applications

Created date

11/15/2022, 4:03:09 PM

Created by

User

Excluded identities (0)

None selected

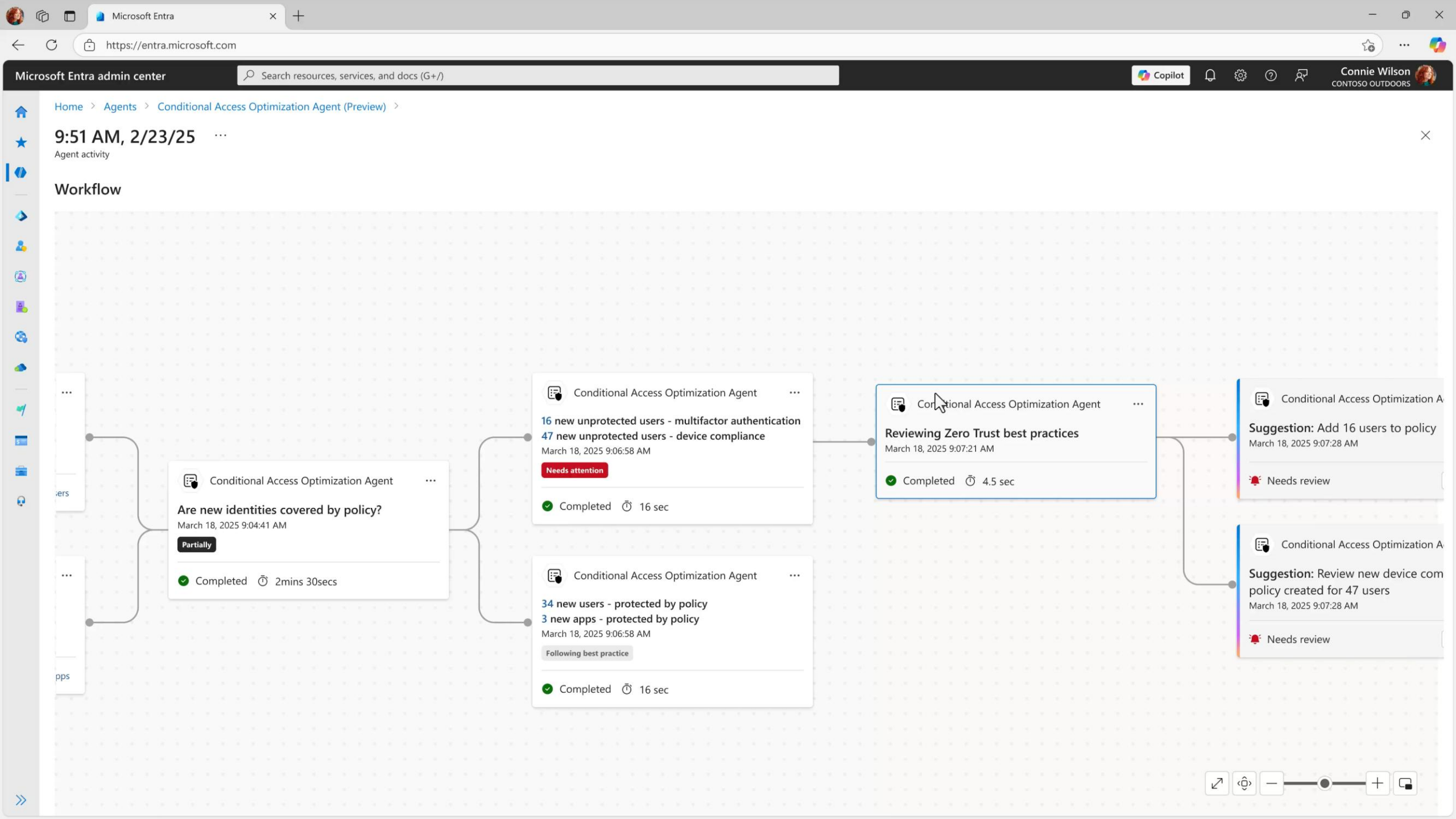
Edit

Requirements for access (1)

Require multifactor authentication

Modified date

6/27/2024, 9:55:36 AM



Microsoft Entra

https://entra.microsoft.com

Microsoft Entra admin center

Search resources, services, and docs (G+/)

Copilot

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CONTOSO OUTDOORS

Home > Agents > Conditional Access Optimization Agent (Preview) >

9:51 AM, 2/23/25
Agent activity

Workflow

Conditional Access Optimization Agent
Agent run '8:00 AM, 2/23/25' started by trigger
March 18, 2025 9:01:05 AM
Completed 3.4 sec

Conditional Access Optimization Agent
Are there new users?
March 18, 2025 9:01:05 AM
Yes
Completed 45 sec
View new users

Conditional Access Optimization Agent
Are there new applications?
March 18, 2025 9:01:05 AM
Yes
Completed 1 min 10 sec
View new apps

Conditional Access Optimization Agent
Are new identities covered by policy?
March 18, 2025 9:04:41 AM
Partially
Completed 2mins 30secs

16 new
47 new
March
Needs

34 new
3 new
March
Follow

Workflow diagram showing the execution of the Conditional Access Optimization Agent. The workflow starts with an agent run triggered at 8:00 AM on 2/23/25, which then branches into two parallel tasks: 'Are there new users?' and 'Are there new applications?'. Both tasks are completed. The 'Are there new users?' task leads to a summary card showing 16 new users and 47 new users, with a 'Needs' status. The 'Are there new applications?' task leads to a summary card showing 34 new applications and 3 new applications, with a 'Follow' status.

Workflow diagram showing the execution of the Conditional Access Optimization Agent. The workflow starts with an agent run triggered at 8:00 AM on 2/23/25, which then branches into two parallel tasks: 'Are there new users?' and 'Are there new applications?'. Both tasks are completed. The 'Are there new users?' task leads to a summary card showing 16 new users and 47 new users, with a 'Needs' status. The 'Are there new applications?' task leads to a summary card showing 34 new applications and 3 new applications, with a 'Follow' status.

Microsoft Entra admin center

Search resources, services, and docs (G+)

Copilot

Connie Wilson
CONTOSO OUTDOORS

Home > Agents > Conditional Access Optimization Agent (Preview) >

9:51 AM, 2/23/25
Agent activity

Workflow

imization Agent

practices

Conditional Access Optimization Agent

Suggestion: Add 16 users to policy
March 18, 2025 9:07:28 AM

Needs review

Review policy

Conditional Access Optimization Agent

Suggestion: Review new device compliance policy created for 47 users
March 18, 2025 9:07:28 AM

Needs review

Review policy

CA3: MFA for Engineering

Conditional Access policy

Policy detailsPolicy impact (Preview)

EditDuplicateDownload JSONDelete

Agent suggestion: Add 16 users to policy

Conditional Access Optimization Agent

The agent found 16 new users that are not in scope of any Conditional Access policies requiring multifactor authentication (MFA) control. The agent has created a new group for those users and has identified the policy CA3: MFA for Engineering as an existing policy that the group can be added to. This policy require users to satisfy MFA controls before accessing resources targeted in the policy. These new users were added to the Contoso tenant by Lauren Baker on 2/22/25 at 1:56 PM.

Apply suggestionReview policy changes

View agent's full activity

AI-generated content may be incorrect. Check it for accuracy

Policy details

Name

CA3: MFA for Engineering

State

Enabled

Edit

Included identities (1)

1 group

View all

Included cloud apps (1)

All applications

Created date

11/15/2022, 4:03:09 PM

Created by

User

Excluded identities (0)

None selected

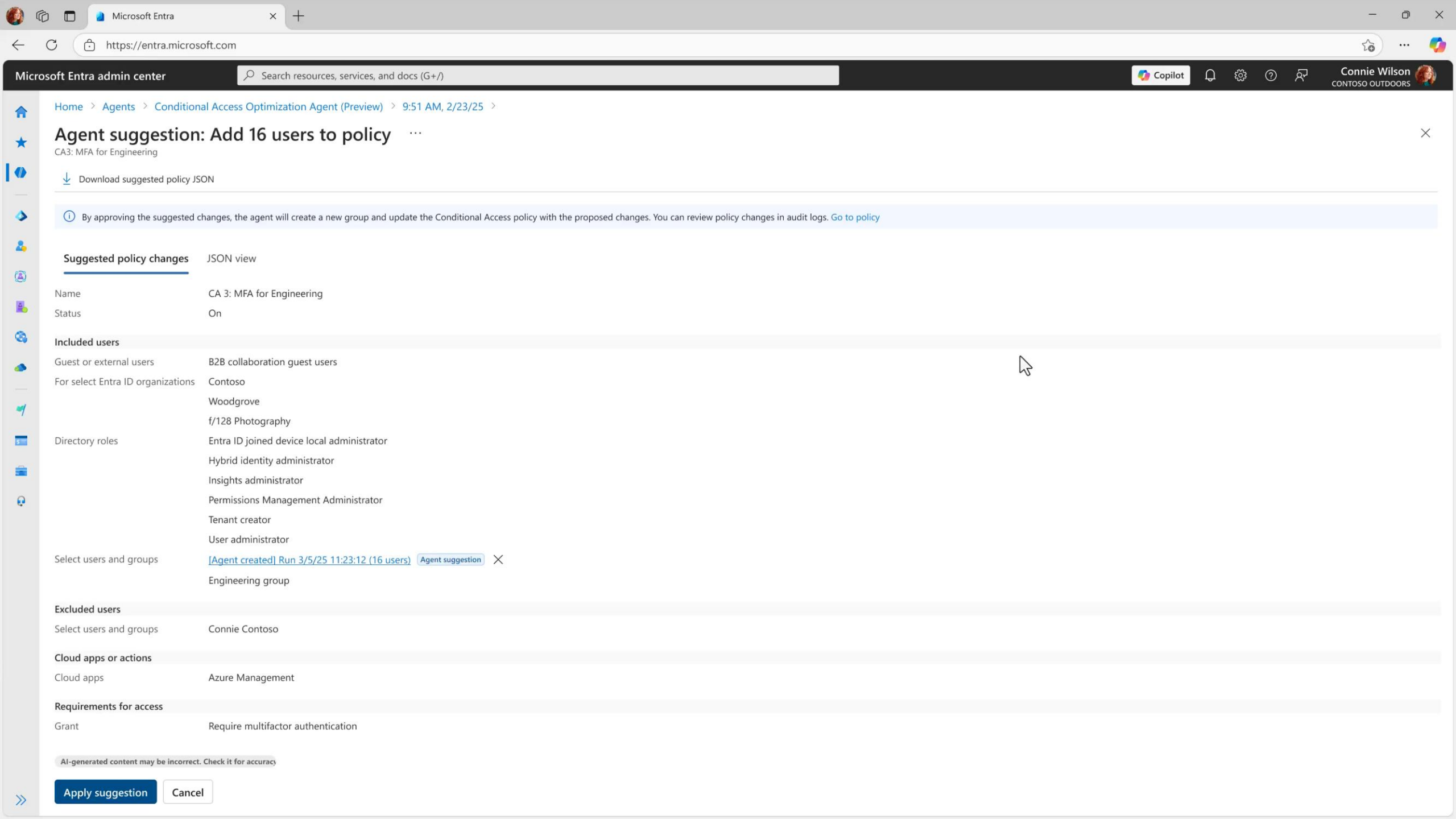
Edit

Requirements for access (1)

Require multifactor authentication

Modified date

6/27/2024, 9:55:36 AM



Directory roles

- Entra ID joined device local administrator
- Hybrid identity administrator
- Insights administrator
- Permissions Management Administrator
- Tenant creator
- User administrator



Select users and groups

[\[Agent created\] Run 3/5/25 11:23:12 \(16 users\)](#)

Agent suggestion

✕

Engineering group

Excluded users

Select users and groups

Connie Contoso

Cloud apps or actions

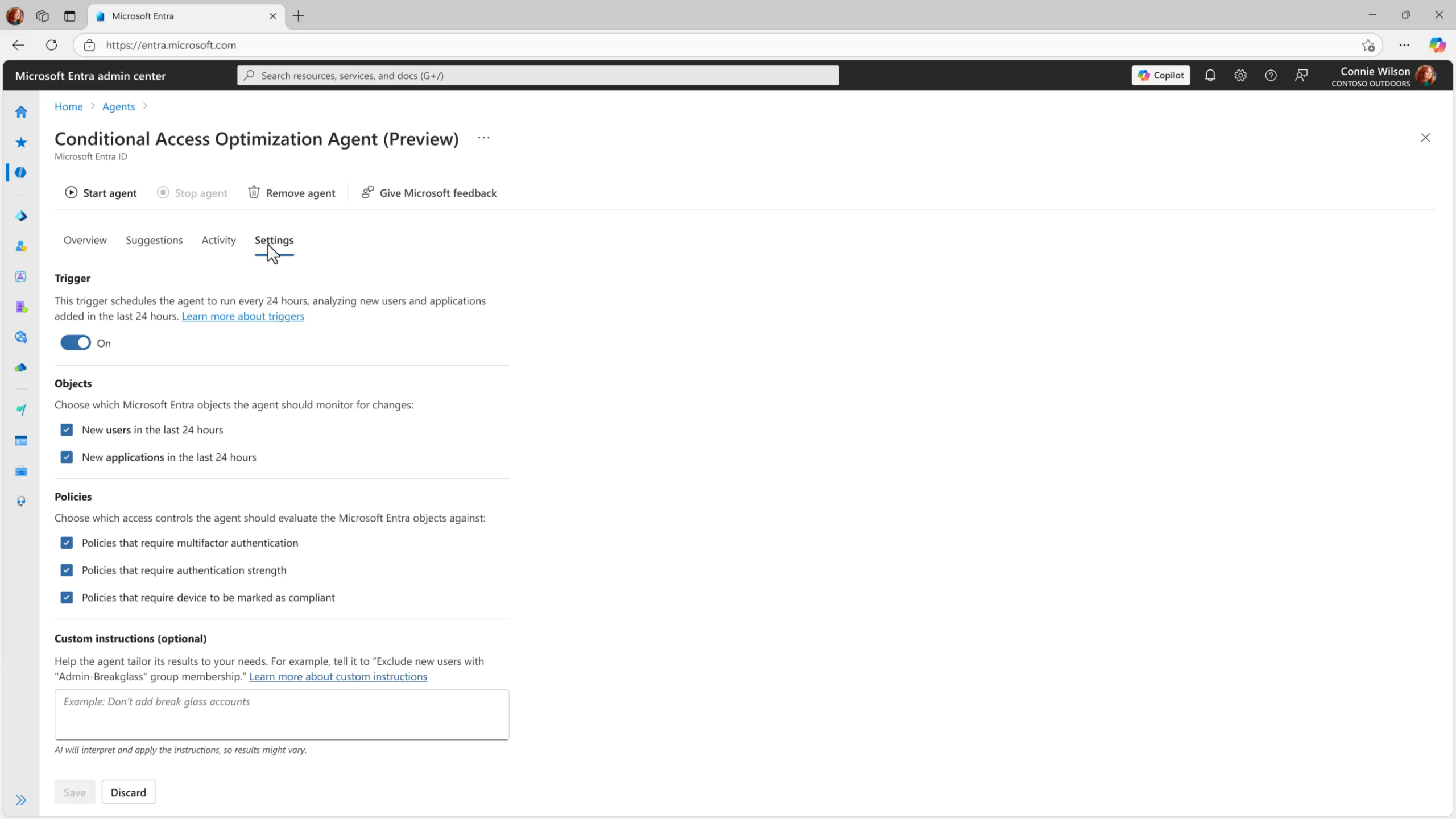
Cloud apps

Azure Management

Requirements for access

Grant

Require multifactor authentication



Key Takeaways

- ...not perfect but takes up rapidly
- ...is here to stay, embrace responsible
- ...ensure a healthy value-cost ratio (use-cases)
- ...have at least one SCU to your disposal
- ...prompt engineering is fundamental for success
- ...respects "OBO" (On-Behalf-Of) principle

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A background image featuring Keanu Reeves in a black turtleneck and sunglasses, looking directly at the camera. The background is a dark green with vertical columns of glowing green text, including letters and numbers, reminiscent of the 'Matrix' digital rain effect. The Microsoft Copilot logo is faintly visible in the center background.

Useful resources

<https://github.com/Azure/Security-Copilot>

<https://github.com/JanVidarElven/copilot-for-security-plugins>

<https://rogierdijkman.medium.com/part-1-develop-a-basic-security-copilot-plugin-bb2d317f76ef>

<https://learn.microsoft.com/en-us/copilot/security/prompting-security-copilot>

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